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The effect of venture capitalists straying from their industry comfort zones

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ABSTRACT

I empirically test and find evidence that venture capital industry experience is more informative and impactful in venture capital investment when measured at the low non-aggregated industry level. Additional evidence shows that venture capital investments made outside of a venture capitalist's preferred investment industry suffer from significant decreases in exit success likelihood. This effect becomes stronger the more distant the investment industry is from the venture capitalist's preferred investment industry.

1. Introduction

According to Amit et al. (1998), the principal role of a venture capital firm(VCF hereafter) is to reduce the cost of informational asymmetries that exist for entrepreneurial firms. For this reason it is unsurprising that VCFs are known for developing specializations (e.g., Dimov and De Clercq, 2006; Hochberg et al. (2015); Siddiqui et al., 2016). Some studies have shown gains associated with the amount of experience a VCF acquires in a particular industry (Gompers et al., 2009; Sorensen, 2009). However, a large limitation in the literature so far has been that past experience has only been measured in a few broad industry groupings. I contribute to reducing this limitation by using detailed and tiered industry groups, which allows me to measure VCF industry experience in both preferred and non-preferred industries.

Most literature has used small datasets and has grouped firms into six broad industry categories. My large international dataset (208,079 observations) splits these same six industries into 69 low-level industry groups. These narrow industry definitions allow a much clearer picture of industry experience. Furthermore, the industries in my setting are defined in a tiered structure, allowing for clear definitions of industry relatedness not influenced by VCF investing patterns or trends. Using these measures of industry relatedness, I test the effect of industry experience on investment outcomes, when this experience is from the same, adjacent, tangential, or even completely unrelated industries. To my knowledge this is the first study to explore the effect of industry experience at such a well populated and detailed industry level. Additionally, this is the first study to jointly consider the effect of industry experience and industry proximity on investment outcomes. Results of these tests are particularly relevant to entrepreneurs when deciding among venture capital financiers.

My results contribute novel findings to the literature by: (1) highlighting the importance of measuring experience at the lowest industry level; (2) documenting the frequency and effect of non-preferred industry investment, both on exit success and on industry-specific experience; and (3) showcasing the effect of industry experience adjacency on exit outcomes.

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2. Theoretical framework and hypothesis development

Many papers have provided evidence that there are gains to specialization in the field of venture capital and/or private equity (Gompers et al., 2009; Sorensen, 2009; Cressy et al., 2007). One strand of this research is the work on style drift, which is when investors begin to make investments outside their originally intended investment purview, typically attempting to find higher returns (see: Cumming et al., 2009).[1] Bubna et al. (2013) find that VCFs that experience style drift are less likely to have successful investments. Consistent with this Rigaonti et al. (2016), show that industry and stage specialization improves exit performance for private equity investment. Related literature has suggested that one potential cost of style drift is greater future fund raising difficulties (Loos and Schwetzler, 2017; and Lahr and Trombley, 2020). The literature so far has clearly shown gains to specialization and costs to straying from known specialties, but has only done so using largely aggregated data.

The majority of prior literature (e.g., Gompers, et al., 2009; Sorensen, 2009) largely ignores more specific or low-level industry groupings and consistently groups together sub-industries, resulting in just 6 to 10 broad industries for the entire economy. While the grouping of sub-industries is helpful to ensure each industry is well populated, it comes at the cost of creating industry groups with less comparability between firms. Thus, understating relevant industry-specific experience, overstating adjacent industry experience, and ignoring the relative adjacency of VCFs' past experience. From the accounting literature, Alford (1992) suggests that the use of less aggregated industry classifications leads to greater comparability between jointly classified firms. With this in mind, my first hypotheses (**H1**) is that measuring industry experience at a less aggregated industry level will make a superior industry experience measure that will be more relevant to explaining investment success than more aggregated measures.

A VCF's acquisition of knowledge in a particular area should result in a form of specialization and should improve future related performance.[2] Consistent with this idea, Dimov and De Holan (2010) show that the less familiar an industry is to a VCF, the less likely the VCF is to enter that industry. Given the many documented gains associated with specialization, and the hesitancy of VCFs to invest in unknown industries, my second hypothesis (**H2**) is that VCFs investing outside of their preferred industry of investment (non-preferred investment industries), versus investment within their preferred industry, will have significantly lower likelihoods of exit success. Although this concept has been test previously, it has never been tested if this remains true at the fine detailed industry level.[3]

Sorensen (2009) never tested the premise, but suggested that experience within a particular industry may have some relevance for adjacent industries (those that would be within the same industry using a broader industry definition). This suggests that the more relatively distant an industry is from a VCF's preferred investment industry, the less relevant the VCF's preferred industry experience becomes. Thus my third hypothesis (**H3**) is that the greater the difference between the VCF's preferred investment industry and the industry of the portfolio company, the more pronounced the non-preferred investment industry effect will be.

3. Research method

3.1. Data and sample

I draw my sample of venture capital investments from the VentureXpert database over the period from 1986 to 2014.[4] Observations are limited to deals identified as being venture capital deals, and where the investing VCF has had at least 10 prior investments.[5] The final sample has 208,079 VCF-round specific investments in 77 different countries. The sample contains a total of 10,435 unique VCFs, 134,386 unique investment rounds, and a total of 59,657 unique portfolio companies. All observations in my study are at the unique VCF-round level.

I make use of the four distinguishable levels of industry consolidation created by the data vendor. I call the most consolidated industry level the broad industry group, which consists of the following six groups: Biotechnology, Communications and media, Computer related, Medical/health/life science, Semiconductors/electrical, and Non-high-technology. These six broad industry groups can be broken down into 10 high-level industries, 18 mid-level industries, and 69 low-level industries. Key to my study is identifying a

[1] A working paper, Hull (2019), provides evidence that VCFs investing outside of their developed specialization primarily due to a lack of preferred industry deal flow and not due to efforts to diversify.

[2] Fulghieri and Sevilir (2009), Sorensen (2009), Gompers et al. (2009), and others support the notion that there are gains to specialization.

[3] As an example, prior studies such as Fulghieri and Sevilir (2009) and Sorensen (2009) suggest a performance loss when a VCF invests in a different broad industry group, such as an investment in the medical/health/life science industry when their most commonly invested in broad industry is the computer related industry. What I document is the effect on success when a VCF invests in the computer programing low-level industry when the VCF's preferred low-level industry is computer software. Where both of these low-level industries are within the same mid-level, high-level and broad industry groups. Figuratively measuring differences between investment industries not with kilometers (broad industry groups) but with millimeters (low-level industry groups).

[4] Data are scarce prior to 1986. Data are also available after 2014 but, to limit the effect of truncation bias, I allow firms at least two years to exit; results are unchanged if five years to exit are given. Additional unreported results show that results remain consistent even allowing firms 15 years to exit their investments. Please refer to the online internet appendix for more information on the selected sample and all variables created.

[5] Results are unchanged without this limitation. This limitation is made as my focus on investment preferences and so prior investment history needs to be observed before an investment can be distinguished between a preferred or non-preferred industry investment.

VCF's preferred investment industry, which in this study is consider to be the low-level industry that the VCF has most commonly invested in at the date of the current observation.[6] Data on exit outcomes comes from the VentureXpert database, which I supplement with exit data from SDC's IPO and M&A databases. Lastly, I gathered country level data from the world bank website and from the *CIA World Factbook*.

3.2. Measures

3.2.1. Dependent variable

I use *Exit success* as my main outcome variable. Following the literature (e.g., Chemmanur et al., 2016; Sorensen, 2009) I define *Exit success* as equal to one if the investment leads to an exit via an IPO, a merger, or an acquisition, and zero otherwise.[7] Although the two different types of exits may be motivated by different reasons, the commonly held belief in the literature is that both are accepted proxies of success. I also follow the literature (e.g., Sorensen (2009), Chemmanur et al. (2016), Gompers et al. (2009), treating investments that failed to exit as unsuccessful investments. Several robustness checks were preformed to ensure results are not sensitive to how exit success is defined.[8]

3.2.2. Independent variable of interest

I consider a VCF's preferred industry to be the low-level industry most commonly invested in by the VCF.[9] I use preferred industry investment as the base case; as such, the main variable of interest will be the dummy variable, *Non-pref. ind.*, which is equal to one if an investment is outside the VCF's preferred investment industry and is zero otherwise.[10] It is worth noting that *Non-pref. ind.* is a backward looking variable, which also entails that a firm can change its industry preference over time.[11]

3.2.3. Control variables

I also use the following standard control variables: *Log VCF exp.*, which is the natural log of the total number of past deals a VCF has invested in; *Round number*, the current financing round number; *Number of VCFs*, the number of venture capital firms investing in the portfolio company (firm receiving venture capital financing); and *Log deal amount*, the total amount of venture financing received by the portfolio company for a given investment round. I additionally control for: the log of GDP of the home country of the portfolio company; the log of stock market capitalization of the home country of the portfolio company; portfolio company country fixed effects to control for country-specific characteristics such as legal structure (see, e.g., La Porta et al., 1997, 1998); investment year fixed effects; broad industry fixed effects; and venture capital deal development stage (i.e., early, late, startup/seed, expansion, buyout/acquisition, and other).[12]

3.3. Analysis technique

I conduct OLS regressions on exit rates of venture capital backed firms through initial public offerings (IPOs) and acquisitions.[13] The baseline regression model is:

$$Exit\ success = \alpha + \beta_1 Non - pref.\ ind._{iv} + \beta_2 X_i + \beta_3 Y_{iv} + \gamma_c + \delta_j + \rho_t + e_{iv.} \tag{1}$$

Here, i indexes the venture capital deal round, v indexes for the unique VCF involved in the deal, j indexes industry, c indexes the country, and t indexes year. As described before, *Exit success* is a dummy variable that is equal to one if the portfolio company has an eventual successful IPO or merger, and zero otherwise; *Non-pref. Ind.*$_{iv}$ refers to the dummy for an investment in a deal that is outside the preferred investment industry of the investing VCF; X_i are deal varying control variables; Y_{iv} are VCF and deal specific control variables; γ_c are country of the portfolio company fixed effects; δ_j are broad industry fixed effects; and ρ_t are the investment year fixed

[6] Low-level industry is the industry most frequently invested in by VCFs, when considering all investments prior to the current investment. As a robustness check, the VCF's preferred industry and experience measures were also created based only on the most recent 5 or 10 years before the current observation. These tests yield similar results and can be found in the online internet appendix.

[7] One potential concern is that the exit outcome variable does not vary at the VCF-round level, potentially biasing my results. To mitigate this concern in section 3.3 of the internet appendix I repeat the paper's analysis using only one observation per portfolio firm. Results of these test produce similar results, alleviating this potential concern.

[8] Please refer to section 4.2 of the online internet appendix for details on these robustness checks.

[9] Results are similar if a VCF's preferred industry is determined using only the most recent 5 or 10 years before the current observation. These test can be found in the online internet appendix.

[10] Using each VCF's own preferred industry investment as the base case is important, as this ensures that my results reflect the most relevant opportunity cost that the VCF faces when making alternative investments.

[11] It is worth mentioning that the data vendor also provides data on a VCF's so called 'industry preference'; however, these data were not used for the main tests of the paper, as this measure is a forward looking measure that does not vary over time. As a robustness check, all tests have been repeated using the industry preferences indicated by the data vendor and results remain qualitatively similar.

[12] Results remain virtually unchanged if VCF-specific fixed effects are used or if a VCF's network size is controlled for in all regressions.

[13] Results are identical if logistical regressions are used. OLS regressions were used due to the concern that logistical regressions with many fixed effects may result in biased estimations. I am grateful to an anonymous referee for the suggestion.

effects. I additionally cluster standard errors at the portfolio company country level.[14]

4. Results

4.1. Summary statistics

Table 1, Panel A provides summary statistics for the main variable of interest *Non-pref. ind.* and for the control variables. The summary statistics for *Non-pref. ind.* show that VCFs invest outside of their narrowly defined preferred investment industry 75.33 % of the time.[15] Additionally, I create a measure of the degree to which an investment preference is different from the industry of the portfolio company (*Degree of industry diff.*), using the different industry consolidation levels.[16] This variable is the number of levels of industry consolidation needed before the two industries are in the same industry group. This variable is equal to zero when the low-level industry matches, one when the mid-level industry match (but not the lowest level), two when the high-level industries match (but not any lower levels), three when the broad industry groups match (but not any lower levels), and four when the industries are in different broad industry groups.[17]

The summary statistics for *Partner w/ind. pref* suggest that about 14 % of the sample have a partner firm that specializes in the industry of the portfolio company. Panel B of Table 1 shows the univariate test for exit success as influenced by *Non-pref. ind.*, indicating that VCFs have a lower likelihood of exit success when investing outside of their investment preference.

4.2. Measuring industry experience at different levels of industry consolidation

For all the regressions, I control for the investing VCF's total experience at the time of the investment, and in Columns 2–5 of Table 2, I additionally control for different industry-specific experience levels from the broad industry level to the most specific low-level industry. Prior literature often measures industry experience using only six industries; this entails the variable *Log VCF exp. in broad industry,* used in Column 2, can be thought of as a proxy for the industry experience measure used in other papers. The results show, first, the significant impact that industry-specific experience has on exit outcomes, and, second, that the more narrowly defined an industry becomes, the more impactful (the greater the coefficient) industry experience is for exit success likelihoods. In this setting measuring low-level industry experience versus broad industry experience is associated with a 33 % increase in coefficient size. Furthermore, Column 6 shows that, when all levels of industry experience are simultaneously controlled for, only low-level industry-specific experience is statistically significant. This suggests that low-level industry-specific experience was what was driving the positive significant coefficients in Columns 2–4.[18] These results show support for **H1**, that the more specific the measure of industry experience, the more applicable the variable is for predicting the success of the current investment.

4.3. Effect of investing outside of a VCF's preferred investment industry

In Table 3 the effect of non-preferred industry investments are explored. In Column 1, I observe a negative relationship between VCFs investing outside of their industry preference and the exit success likelihood of the portfolio company. This provides support for **H2**, suggesting that specialization is of primary importance and that VCFs will have lower likelihoods of exit success when investing outside of their preferred investment industry. The results indicate that VCFs that invest outside of their preferred investment area are significantly less likely to have a successful exit.[19] Co-investing with a VCF that has an industry preference to invest in the industry of the portfolio company does not appear to significantly impact exit success.

Column 2 attempts to recreate the results of Column 1 but now using a non-preferred broad industry investment dummy (*Non-pref. broad ind.*). To be consistent with the new non-preferred broad industry, dummy *Log VCF exp. in broad industry* is used as the measure of industry experience. Column 2 shows that broad industry experience has a positive impact on exit success, but the *Non-pref. broad ind.* dummy is not statistically significant. This indicates that the costs of not investing in a VCF's preferred industry cannot be observed when using aggregated industry data and further indicates the benefit and contribution of measuring industry experience at the low industry level, providing further support to **H1**.

In Columns 3–5, I address the question of whether the effect of investing outside the VCF's preferred investment industry is more

[14] My results are not sensitive to not clustering or to more detailed levels of clustering.

[15] This is not solely driven by the narrow low-level industry definition, as, even using the six broad industry definition, non-preferred industry investment would occur more than 50 % of the time.

[16] Please see Fig. 1 for a more in-depth example of how the degree of industry difference is calculated.

[17] More than 99 % of VC make a non-preferred industry investment at some point in time suggesting that this is not an action only taken by lower quality VCFs. Likewise, these investments are not all clustered during one particular period of time. Analyzing the time between any investment observation and another investment made by the same VCF, but within a non-preferred investment industry, I find the median number of days is 16, with more than 85 % of observations having a non-preferred industry investment within three months.

[18] No difference in results is seen if orthogonalized industry level experience measures are used instead of regular industry level experience measures.

[19] Similar formatted logit regressions indicate that this these investments are 11 % less likely to have a successful exit, than preferred industry investments.

Table 1
Summary statistics. This table reports the summary statistics for the main variables of interest in **Panel A**. In **Panel B** univariate tests between *Non-pref. ind.* and Exit success are performed.

Panel A		(1)	(2)
		Mean	Median
Non-pref. ind.		75.27%	
Degree of industry diff. =0		24.73%	
Degree of industry diff. =1		5.98%	
Degree of industry diff. =2		0.48%	
Degree of industry diff. =3		16.43%	
Degree of industry diff. =4		52.38%	
Partner w/observed ind. pref.		14.93%	
Exp. in 1st tier specialty		27.39%	24.48%
Exp. in 2nd tier specialty		14.52%	13.33%
Exp. in 3rd tier specialty		9.43%	10.15%
VCF exp. in low-level industry		22.49	6
VCF exp. in mid-level industry		40.58	15
VCF exp. in high-level industry		41.41	16
VCF exp. in broad industry		68.56	25
Total VCF exp.		246.19	103
Round number		3.65	3
Number of VCFs		4.00	3
Deal amount(Mil.)		11409.63	5300
Market cap.(Bil.)		11524.97	13983.67
GDP(Bil.)		9622.44	10284.78
Exit success		41.90%	
Observations		208,079	
Unique VC firms		10,435	
Unique rounds of investment		134,387	
Unique portfolio companies		59,657	
Exit success of unique portfolio companies		29.21%	
Panel B			Exit success
Non-pref. ind.	=0		43.85%
	=1		41.26%
	Diff.		2.59***
	Observations		208,079

severe the greater the difference between the investment industry and the VCF's preferred investment industry. It is worth mentioning that this analysis is not possible using broad industry group experience, as all investments are either inside or outside of a broad industry group and so no measure of adjacency between industries can be created. For these columns, the main variable of interest is the *Degree of industry diff*, which is the number consolidation levels required before the VCF's preferred industry and the industry of the portfolio company are in the same industry. In Column 3 the sample is limited to investments that were made outside of a VCF's preferred investment industry. The negative coefficient on the *Degree of industry diff* variable indicates that even among only non-preferred industry investments, the larger the degree of difference between the VCFs preferred industry and the portfolio company's industry, the larger the negative effect. In Column 4 I use the full sample and show the same supportive results. Column 5 shows both effects at once, namely the drop in exit likelihood for a non-preferred industry investment and an additional drop in exit likelihood the more distant the investment industry is from the VCF's preferred investment industry. It worth noting that there is a non-linearity in the effect of investing in a non-preferred investment industry, with a large drop in success rates moving from the preferred to an adjacent industry and a small drop in success moving from an adjacent to a more distant industry. The results in Table 3 show support for **H3**, the hypothesis that the greater the difference between the VCF's investment preference and the industry of the portfolio company, the more pronounced the non-preferred investment effect will be.[20]

4.4. Robustness checks

In the online internet appendix a battery of robustness checks are performed to support the above findings. Results are shown to not be due to high risk investments or to other well documented venture capital control variables.[21] Results are shown to be robust to

[20] These results provide additional support for **H2,** that VCFs have lower likelihoods of exit success when investing outside of their preferred investment industry.
[21] Buchner et al. (2017) suggest that diverse industry investments could be the motivated by a high risk strategy, while Nahata, Hazarika & Tandon (2014) and Tykvova (2018) suggest many possible venture capital investment controls.

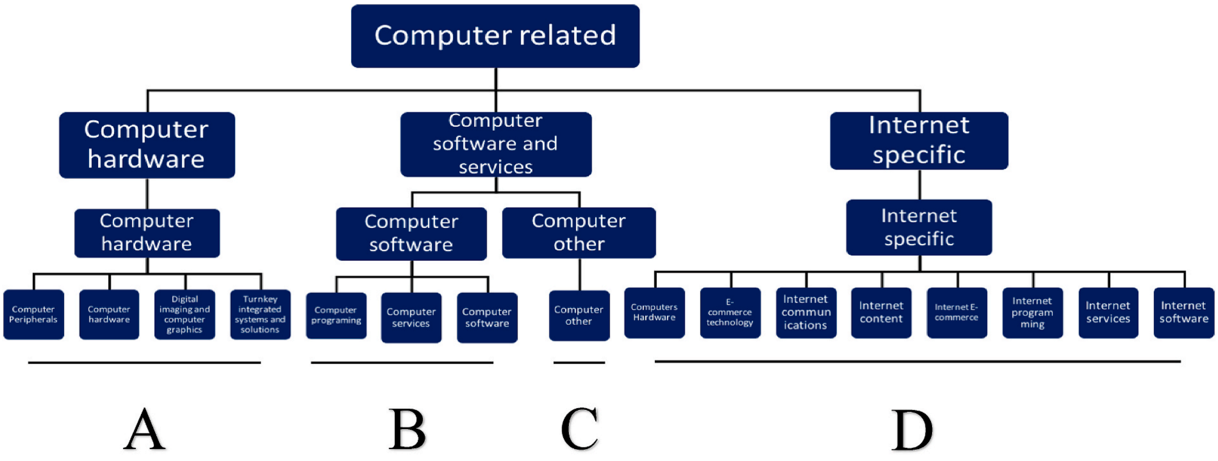

Fig. 1. Example of 4 levels of industry consolidation. The diagram below depicts as an illustrative example of the four levels of industry consolidation that can be seen under one of the broad industry groupings, that of the computer related industry. Each of the smallest boxes represents its own unique low-level industry. As an example please consider a VCF with the preferred industry of Computer services (within group B). If this VCF invests in a deal also within the Computer services industry this would be within the VCF's preferred industry and so the industry difference would be 0. Any investment within group B, but outside of the computer service industry, would be considered 1 degree of industry difference, as the two different industries have the same mid-level industry. Likewise, an investment in a group C industry would be considered 2 degrees of industry difference, as the two different industries share the same high-level industry. An investment in a group A or D industry would be considered 3 degrees of industry difference, as the these different industries share the same broad industry group. Lastly, any investment in a different broad industry group than the Computer related broad industry group would be considered 4 degrees of industry difference as the groups are in unrelated industries. In total, there are 6 broad industry groups, 10 high-level industry groups, 18 mid-level industry groups, and 69 low-level industry groups.

Table 2

The effect of industry aggregation on exit success. OLS regressions where the dependent variable is *Exit success*. Heteroskedasticity-corrected robust standard errors, which are clustered at the country level, are in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10 % levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Exit success	Exit success	Exit success	Exit success	Exit success	Exit success
Log VCF exp. in low-level industry					0.016***	0.016***
					[0.001]	[0.001]
Log VCF exp. in mid-level industry				0.014***		0.007
				[0.002]		[0.010]
Log VCF exp. in high-level industry			0.013***			−0.007
			[0.001]			[0.008]
Log VCF exp. in broad industry		0.012***				−0.000
		[0.001]				[0.003]
Log Total VCF exp.	0.008***	−0.002	−0.003	−0.003	−0.003	−0.002
	[0.001]	[0.001]	[0.002]	[0.003]	[0.002]	[0.002]
Partner w/current ind. pref.	−0.003	−0.001	−0.000	−0.000	−0.003	−0.003
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Round number	0.007***	0.006***	0.006***	0.006***	0.006***	0.006***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Number of VCFs	0.006***	0.006***	0.006***	0.006***	0.006***	0.006***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Log deal amount	0.031***	0.031***	0.031***	0.031***	0.031***	0.031***
	[0.001]	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]
Log market cap.	−0.000***	−0.000***	−0.000***	−0.000***	−0.000***	−0.000***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Log GDP	0.000	0.000	0.000	0.000	0.000	0.000
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Observations	208,079	208,079	208,079	208,079	208,079	208,079
Adjusted R-sq	0.151	0.151	0.152	0.152	0.152	0.152
Investment country FE	Y	Y	Y	Y	Y	Y
Investment Year FE	Y	Y	Y	Y	Y	Y
Major Industry FE	Y	Y	Y	Y	Y	Y
VCF stage FE	Y	Y	Y	Y	Y	Y

Table 3
The effect of VCFs investing outside of their preferred investment industry. OLS regressions where the dependent variable is *Exit success*. Heteroskedasticity-corrected robust standard errors, which are clustered at the country level, are in brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10 % levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Exit success	Exit success	Exit success	Exit success	Exit success
Non-pref. ind.	−0.024***				−0.017***
	[0.003]				[0.004]
Non-pref. broad ind.		−0.001			
		[0.003]			
Degree of industry diff.			−0.005***	−0.006***	−0.002***
			[0.001]	[0.001]	[0.001]
Log VCF exp. in low-level industry	0.010***		0.008***	0.012***	0.010***
	[0.002]		[0.002]	[0.002]	[0.002]
Log VCF exp. in broad industry		0.012***			
		[0.001]			
Log Total VCF exp.	0.005	0.002	0.010**	0.005	0.006
	[0.005]	[0.007]	[0.005]	[0.005]	[0.005]
Partner w/current ind. pref.	0.001	−0.002	0.001	0.001	0.002
	[0.002]	[0.001]	[0.002]	[0.002]	[0.002]
Round number	0.006***	0.006***	0.006***	0.006***	0.006***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Number of VCFs	0.006***	0.006***	0.005***	0.006***	0.006***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Log deal amount	0.031***	0.031***	0.032***	0.030***	0.030***
	[0.002]	[0.001]	[0.002]	[0.002]	[0.002]
Log market cap.	−0.000***	−0.000***	−0.000***	−0.000***	−0.000***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Log GDP	0.000	0.000	0.000	0.000	0.000
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Observations	208,079	208,079	156,631	208,079	208,079
Pseudo R-sq	0.153	0.151	0.145	0.152	0.153
Investment country FE	Y	Y	Y	Y	Y
Investment Year FE	Y	Y	Y	Y	Y
Major Industry FE	Y	Y	Y	Y	Y
VCF stage FE	Y	Y	Y	Y	Y

alternative ways of measuring investment experience and hold in restricted sample tests. Results are also shown not to be sensitive to investment complexity or alternate VCF's preferred investment industry definitions. Additionally, all regressions are found to be robust to the inclusion of VCF fixed effects, effectively comparing preferred and non-preferred industry investments for the same VCF. Additional firm matching tests and 2 types of endogeneity corrections are also contained in the internet appendix and provide further support of the above document results.

5. Conclusion

This research contributes some important findings to the literature. First, by providing evidence that VCFs' industry-specific experience is more meaningful when defined at the low-industry level rather than the broad industry level. These findings give greater insight into the extent that venture capitalist specialize, even to narrowly defined low-industry classification level. This suggest that prior utilized measures of broad industry expertise have likely not fully captured the true effect of specific industry expertise or related industry expertise.

Second, this research adds empirical evidence on the transferability of experience to related industries. Experience is shown to be only partially transferable between industries and the ease by which this experience transfers is directly related to the degree of relatedness between the VCF's preferred investment industry and the industry of the portfolio company in a non-liner pattern.

Another important contribution to the literature is on the methodology of comparing the effectiveness of different investing patterns. Prior literature implicitly compares investment outcomes to the average outcome of the VCF. While in this study, I benchmark exit outcomes by the average likelihood of exit success in the VCF's preferred low-level investment industry, thus framing all non-preferred investments in terms of their appropriate opportunity cost.

From a policy perspective, my findings suggest that certain investment industries could be overlooked by VCFs that have not developed an expertise for working in a particular industry. Governments that wish to promote the growth of a particular industry should incentivize VCFs to make multiple investments in a particular industry, thus encouraging VCFs to acquire a new specialty that will lead to subsequent investments within the industry by the investing VCF.

From an entrepreneur's perspective, entrepreneurs should seek venture capital backing from VCFs specialized in the entrepreneurs' specific industry, even if it means partnering with a less experienced VCF.

Overall, my results showcase the importance of industry specialization and that this specialization occurs at the non-aggregated industry level. This is shown by large decreases in exit success likelihoods for VCF investments in non-preferred industries. Furthermore, these costs are shown to be more severe the less adjacent the industry is to the VCF's preferred investment industry.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbvi.2021.e00266.

Credit author statement

Tyler Hull: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing- Original Draft, Writing- Review & Editing and Visualization.

References

Alford, A., 1992. The effect of the set of comparable firms on the accuracy of the price-earnings valuation method,. J. Account. Res. 30, 94–108. http://hdl.handle.net/10.2307/2491093.

Amit, R., Brander, J., Zott, C., 1998. Why do venture capital firms exist? Theory and Canadian evidence. J. Bus. Ventur. 13 (6), 441–466. https://doi.org/10.1016/S0883-9026(97)00061-X.

Bubna, A., Das, S., Hanouna, P., 2013. Persistence pays: evidence from investment style dynamics in the venture capital industry. Working paper. Indian School of Business. https://doi.org/10.2139/ssrn.2286213. Santa Clara University, and Villanova University.

Buchner, A., Mohamed, A., Schwienbacher, A., 2017. Diversification, risk, and returns in venture capital. J. Bus. Ventur. 32, 519–535. https://doi.org/10.1016/j.jbusvent.2017.05.005.

Chemmanur, T.J., Hull, T., Krishnan, K., 2016. Do local and international venture capitalists play well together? The complementarity of local and international venture capitalists, Journal of Business Venturing, 31-5,573-594. https://doi.org/10.1016/j.jbusvent.2016.07.002.

Cressy, R., Munari, F., Malipiero, A., 2007. Playing to their strengths? Evidence that specialization in the private equity industry confers competitive advantage. J. Corp. Finance 13 (4), 647–669. https://doi.org/10.1016/j.jcorpfin.2007.04.007.

Cumming, D., Fleming, G., Schwienbacher, A., 2009. Style drift in private equity. J. Bus. Finance Account. 36, 645–678. https://doi.org/10.1111/j.1468-5957.2009.02137.x.

Dimov, D., De Clercq, D., 2006. Venture capital investment strategy and portfolio failure rate: a longitudinal study. Enterpren. Theor. Pract. 30 (2), 207–223. https://doi.org/10.1111/j.1540-6520.2006.00118.x.

Dimov, D., De Holan, P., 1962–2004. 2010, firm experience and market entry by venture capital firms. J. Manag. Stud. 47 (1), 130–161. https://doi.org/10.1111/j.1467-6486.2009.00869.x.

Fulghieri, P., Sevilir, M., 2009. Size and focus of a venture capitalist's portfolio. Rev. Financ. Stud. 22, 4643–4680. https://doi.org/10.2139/ssrn.686172.

Gompers, P., Kovner, A., Lerner, J., 2009. Specialization and success: evidence from venture capital. J. Econ. Manag. Strat. 18 (3), 817–844. https://doi.org/10.1111/j.1530-9134.2009.00230.x.

Hochberg, Y., Mazzeo, M., McDevitt, R., 2015. Specialization and competition in the venture capital industry. Rev. Ind. Organ. 46 (4), 323–347. https://doi.org/10.1007/s11151-015-9462-3.

Hull, T., 2019. Why venture capitalists stray from their industry comfort zones. Working Paper. University of Massachusetts-Boston. https://doi.org/10.2139/ssrn.3303466.

La Porta, R., Lopez-De-Silanes, F., Shleifer, A., Vishny, R., 1997. Legal determinants of external finance. J. Finance 52, 1131–1150. https://doi.org/10.1111/j.1540-6261.1997.tb02727.x.

La Porta, R., Lopez-De-Silanes, F., Shleifer, A., Vishny, R., 1998. Law and finance. J. Polit. Econ. 106, 1113–1155. https://doi.org/10.1086/250042.

Lahr, Henry, Trombley, Timothy E., 2020. Early indicators of fundraising success by venture capital firms. J. Corp. Finance 65. https://doi.org/10.1016/j.jcorpfin.2020.101672.

Loos, R., Schwetzler, B., 2017. Fueling the buyout machine: fundraising in private equity. Financ. Mark. Portfolio Manag. 31 (4), 397–443. https://doi.org/10.1007/s11408-017-0298-8.

Nahata, R., Hazarika, S., Tandon, K., 2014. Success in global venture capital investing: do institutional and cultural differences matter? J. Financ. Quant. Anal. 49 (4) https://doi.org/10.1017/S0022109014000568.

Rigamonti, D., Cefis, E., Meoli, M., Vismara, S., 2016. The effects of the specialization of private equity firms on their exit strategy,. J. Bus. Finance Account. 43, 9–10. https://doi.org/10.1111/jbfa.12221, 1420-1443.

Siddiqui, A., Marinova, D., Hossain, A., 2016. Venture capital firms' specialization, differences and complementarities. Int. J. Bus. Manag. 11 (83) https://doi.org/10.2139/ssrn.2571192.

Sorensen, M., 2009. Learning by investing: evidence from venture capital. Working Paper. Columbia University.

Tykvova, T., 2018. Legal framework quality and success of (different types of) venture capital investments. J. Bank. Finance 87, 333–350. https://doi.org/10.1016/j.jbankfin.2017.10.015.